Federated Learning for Wildlife Monitoring: Privacy-Preserving Multi-Site Analysis

Cross-Institutional Data Collaboration with Differential Privacy and Heterogeneous Sensor Fusion

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RESEARCH

Abstract

Wildlife monitoring generates vast quantities of sensitive data across geographically distributed sites managed by independent institutions with heterogeneous sensor deployments and conflicting data-sharing policies. Training unified machine learning models on this fragmented landscape traditionally requires centralizing raw data—a process that is often legally prohibited, ethically problematic, and logistically impractical. We present ZooFed, a federated learning framework purpose-built for cross-institutional wildlife monitoring that enables collaborative model training without sharing raw sensor data. ZooFed introduces three innovations: a *Heterogeneous Sensor Adapter* (HSA) that normalizes inputs from camera traps, acoustic recorders, GPS collars, eDNA samplers, and satellite imagery into a shared feature space compatible with federated aggregation; a *Conservation-Calibrated Differential Privacy* (CCDP) mechanism that allocates privacy budgets based on species sensitivity rather than uniform noise injection, providing stronger protection for endangered species locations; and a *Non-IID Robust Aggregation* (NIRA) algorithm that handles the extreme statistical heterogeneity of ecological data across sites with different species assemblages, climatic conditions, and sensor configurations. We evaluate ZooFed on a consortium of 18 wildlife monitoring sites across 9 countries, encompassing 4.2 million sensor events from 847 species. ZooFed achieves 89.4% species detection accuracy, within 2.1 percentage points of a centralized oracle with full data access, while providing (ϵ, δ)-differential privacy guarantees with $\epsilon = 2.0$ for common species and $\epsilon = 0.5$ for critically endangered species. Compared to local-only models, ZooFed improves detection accuracy by 18.7% on average and by 31.2% for rare species that benefit most from cross-site knowledge transfer.

Keywords: federated learning, wildlife monitoring, differential privacy, sensor fusion, cross-institutional collaboration, conservation AI

1 Introduction

Wildlife monitoring has entered the era of pervasive sensing. Camera trap networks capture billions of images annually [Ahumada et al., 2020]. Passive acoustic monitors record continuous soundscapes across terrestrial and marine environments [Gibb et al., 2019]. GPS and satellite telemetry tracks thousands of individual animals [Kays et al., 2015]. Environmental DNA sampling detects species presence from water and soil samples [Deiner et al., 2017]. Collectively, these sensor systems generate an unprecedented volume of ecological data distributed across hundreds of independent institutions.

The potential of this data for biodiversity science and conservation is immense: unified models trained on multi-site data could detect species more accurately, identify population trends earlier, and predict conservation threats more reliably than any single-site model. Yet realizing this potential is blocked by three barriers.

First, *legal and regulatory constraints*. Location data for endangered species is often legally protected; sharing precise coordinates of critically endangered species nesting sites could enable poaching [Lindenmayer et al., 2017]. Data sovereignty regulations increasingly restrict cross-border transfer of environmental data [Fritz et al., 2019].

Second, *institutional incentives.* Research groups invest substantial resources in data collection and are understandably reluctant to share raw data before publishing their analyses [Tenopir et al., 2011]. Competitive funding structures further discourage open sharing.

Third, *technical heterogeneity.* Different sites deploy different sensor types, configurations, and schedules. A camera trap network in the Serengeti produces high-resolution daytime images, while one in Borneo produces primarily infrared night images. Acoustic sensors in marine environments capture very different frequency ranges than those deployed in tropical forests. Training a unified model requires handling this heterogeneity without centralizing the raw data.

Federated learning (FL) [McMahan et al., 2017] offers a promising paradigm: models are trained locally at each site, and only model updates (gradients or parameters) are shared with a central coordinator. Raw data never leaves the originating institution. However, standard FL algorithms face challenges in the wildlife monitoring setting: extreme non-IID data distributions across ecologically diverse sites, heterogeneous sensor modalities, and the need for species-specific privacy guarantees.

We present ZooFed, a federated learning framework designed for the specific challenges of cross-institutional wildlife monitoring. Our contributions are:

1. **Heterogeneous Sensor Adapter (HSA):** A modality-agnostic feature extraction architecture that maps camera trap images, acoustic spectrograms, GPS trajectories, and eDNA reads into a shared embedding space for federated aggregation (Section 4).

2. **Conservation-Calibrated Differential Privacy (CCDP):** A privacy mechanism that allocates noise budgets proportional to species conservation sensitivity, providing stronger protection for endangered species (Section 5).

3. **Non-IID Robust Aggregation (NIRA):** A federated aggregation algorithm that handles extreme distribution heterogeneity across ecological sites through clustered multi-task learning (Section 6).

2 Related Work

2.1 Federated Learning

Federated Learning was formalized by McMahan et al. [2017] with the FedAvg algorithm. Subsequent work addressed non-IID data through personalization [Li et al., 2020], clustering [Ghosh et al., 2020], and multi-task formulations [Smith et al., 2017]. Communication efficiency improvements include gradient compression [Alistarh et al., 2017], sparsification [Stich et al., 2018], and distillation-based approaches [Lin et al., 2020]. Kairouz et al. [2021] provide a comprehensive survey of open problems and advances.

2.2 Differential Privacy in Federated Learning

Differential privacy (DP) provides formal privacy guarantees for statistical analyses [Dwork and Roth, 2014]. Integration with FL was established by McMahan et al. [2018] through user-level DP. Geyer et al. [2017] proposed client-level DP-SGD for federated settings. Adaptive privacy budget allocation has been explored in the context of heterogeneous data sensitivity [Kang et al., 2020], but not for ecological conservation applications.

2.3 Machine Learning for Wildlife Monitoring

Deep learning has been applied extensively to camera trap classification [Norouzzadeh et al., 2018, Beery et al., 2019], acoustic species detection [Kahl et al., 2021, Stowell, 2019], and movement ecology [Patterson et al., 2017]. Multi-modal approaches combining visual and acoustic data have shown improved performance [Verma et al., 2022]. However, most work assumes centralized data access.

Tuia et al. [2022] identified distributed learning as a key frontier for conservation machine learning but noted the absence of domain-specific frameworks. Beery et al. [2021] documented the domain shift problem across camera trap sites, which directly motivates our non-IID robust aggregation approach.

2.4 Privacy in Conservation Data

The sensitivity of conservation data—particularly precise locations of endangered species—has been recognized as a critical concern [Lindenmayer et al., 2017]. Tulloch et al. [2018] analyzed the tradeoff between data openness and conservation risk. Current approaches rely on coarse spatial aggregation (grid-based reporting) rather than formal privacy mechanisms, sacrificing data utility for protection.

3 Problem Formulation

3.1 System Model

We consider a federation of K wildlife monitoring sites, each managed by an independent institution. Site k operates a set of sensors $\mathcal{S}_k = \{s_1^{(k)}, \ldots, s_{n_k}^{(k)}\}$ that collectively produce a local dataset $\mathcal{D}_k = \{(\mathbf{x}_i^{(k)}, y_i^{(k)})\}_{i=1}^{N_k}$ where $\mathbf{x}_i$ is a sensor observation (image, spectrogram, trajectory, or sequence) and y_i is a species label.

The global objective is to train a model f_θ that minimizes the aggregate loss:

$$\min_\theta \sum_{k=1}^{K} \frac{N_k}{N} \mathcal{L}_k(\theta) \quad \text{where} \quad \mathcal{L}_k(\theta) = \frac{1}{N_k} \sum_{i=1}^{N_k} \ell(f_\theta(\mathbf{x}_i^{(k)}), y_i^{(k)}) \tag{1}$$

subject to the constraint that raw data $\mathcal{D}_k$ never leaves site k, and model updates satisfy species-specific differential privacy guarantees.

3.2 Challenges

Three properties of wildlife monitoring data make standard FL insufficient:

Modality Heterogeneity. Different sites deploy different sensor types. Site k_1 may have only camera traps, site k_2 only acoustic sensors, and site k_3 both. Standard FL requires a shared model architecture across all sites.

Extreme Non-IID. Species assemblages vary dramatically across sites. A site in the Amazon may share fewer than 5% of species with a site in East Africa. This creates extreme label distribution skew ($P_k(y) \neq P_j(y)$) and covariate shift ($P_k(\mathbf{x} \mid y) \neq P_j(\mathbf{x} \mid y)$).

Conservation-Sensitive Privacy. Not all data requires the same privacy protection. The location of a critically endangered rhinoceros requires much stronger protection than the location of a common impala. Uniform privacy budgets waste utility by over-protecting common species and under-protecting rare ones.

4 Heterogeneous Sensor Adapter

4.1 Architecture

The HSA architecture decomposes the model into modality-specific encoders and a shared classification head:

$$f_\theta(\mathbf{x}) = h_\phi(g_{\psi_m}(\mathbf{x})) \tag{2}$$

where g_{ψ_m} is the modality-specific encoder for modality m (visual, acoustic, trajectory, genomic) with parameters ψ_m, and h_ϕ is the shared species classification head with parameters ϕ.

Each encoder maps its input to a common embedding space $\mathbb{R}^d$:

$$g_{\psi_m} : \mathcal{X}_m \to \mathbb{R}^d \quad \forall m \in \mathcal{M} \tag{3}$$

Table 1: Modality-specific encoder architectures.

Modality	**Input**	**Architecture**	**Output**
Visual	RGB/IR images	EfficientNet-B0	256-d embedding
Acoustic	Mel spectrograms	ResNet-18	256-d embedding
Trajectory	GPS sequences	Temporal Transformer	256-d embedding
Genomic	eDNA barcodes	1D-CNN + Attention	256-d embedding

4.2 Cross-Modal Alignment

To ensure that embeddings from different modalities are compatible for the shared classifier, we introduce a cross-modal alignment loss during training. For sites with multiple sensor types, we minimize the embedding distance for co-occurring observations:

$$\mathcal{L}_{\text{align}} = \sum_{(m_1,m_2)\in\mathcal{M}^2} \sum_{(\mathbf{x}_{m_1},\mathbf{x}_{m_2})\in\mathcal{C}_{m_1,m_2}} \|g_{\psi_{m_1}}(\mathbf{x}_{m_1}) - g_{\psi_{m_2}}(\mathbf{x}_{m_2})\|^2 \tag{4}$$

where $\mathcal{C}_{m_1,m_2}$ is the set of co-occurring observations from modalities m_1 and m_2 (e.g., a camera trap image and an acoustic recording from the same spatiotemporal window containing the same species).

4.3 Federated Training with HSA

In the federated training protocol, each site trains its local modality encoders and the shared classifier:

1. The coordinator distributes the shared classifier parameters ϕ and the relevant modality encoder parameters ψ_m for each site's sensor types.
2. Each site performs local training on its data, updating both encoder and classifier parameters.
3. Sites send back only the shared classifier gradient $\nabla_\phi \mathcal{L}_k$ to the coordinator.
4. Modality encoder updates $\nabla_{\psi_m} \mathcal{L}_k$ are aggregated only among sites sharing the same modality.

This split allows the classifier to learn from all sites regardless of sensor type, while encoders specialize through modality-specific aggregation.

5 Conservation-Calibrated Differential Privacy

5.1 Motivation

Standard DP-SGD [Abadi et al., 2016] applies uniform noise calibrated to a global privacy budget ϵ. In conservation, this is suboptimal: adding heavy noise to protect rhinoceros locations also degrades the model's ability to classify common species, while light noise for overall utility leaves endangered species insufficiently protected.

5.2 Species-Specific Privacy Budgets

We assign privacy budgets ϵ_s based on species conservation status:

$$\epsilon_s = \epsilon_{\max} \cdot \left(\frac{\text{rank}(s)}{\max_s \text{rank}(s)} \right)^\gamma \tag{5}$$

where $\text{rank}(s)$ maps IUCN Red List categories to ordinal values:

Table 2: Privacy budget allocation by IUCN category.

IUCN Category	**Rank**	ϵ_s ($\gamma = 1$)	ϵ_s ($\gamma = 2$)
Critically Endangered (CR)	1	0.33	0.11
Endangered (EN)	2	0.67	0.44
Vulnerable (VU)	3	1.00	1.00
Near Threatened (NT)	4	1.33	1.78
Least Concern (LC)	5	1.67	2.78
Not Evaluated (NE)	6	2.00	4.00

5.3 CCDP Mechanism

The CCDP mechanism operates during gradient computation at each site:

Algorithm 1 Conservation-Calibrated DP-SGD (per-site, per-round)

Require: Local dataset $\mathcal{D}_k$, model θ, clipping bound C, species budgets $\{\epsilon_s\}$

```
for each minibatch B ⊆ D_k do
  for each example (x_i, y_i) ∈ B do
    Compute per-example gradient g_i = ∇_θ ℓ(f_θ(x_i), y_i)
    Clip: ḡ_i = g_i · min(1, C/‖g_i‖_2)
    Compute noise scale: σ_i = C√(2 ln(1.25/δ)) / ε_{y_i}
    Add noise: g̃_i = ḡ_i + N(0, σ_i^2 I)
  end for
  Aggregate: g̃ = (1/|B|) Σ_i g̃_i
  Update: θ ← θ − η g̃
end for
return Updated model θ
```

5.4 Privacy Accounting

The per-species privacy guarantee follows from the composition theorem for Gaussian mechanisms. After T rounds of training with sampling rate q:

$$\epsilon_s^{\text{total}} \leq q\sqrt{2T\ln(1/\delta)} \cdot \frac{C}{\sigma_s} \tag{6}$$

We use the moments accountant [Abadi et al., 2016] for tighter composition bounds, tracking privacy expenditure separately for each species category and halting training for any species category that exhausts its budget.

5.5 Privacy-Utility Analysis

We analyze the impact of CCDP compared to uniform DP:

Table 3: Privacy-utility comparison: CCDP vs. uniform DP.

Method	**CR/EN Acc. (%)**	**LC Acc. (%)**	**Overall Acc. (%)**	**CR/EN ϵ**
No privacy	84.2	93.1	91.5	∞
Uniform DP ($\epsilon = 1.0$)	71.3	82.4	80.7	1.0
Uniform DP ($\epsilon = 2.0$)	78.1	88.7	87.2	2.0
CCDP ($\gamma = 1$)	79.8	90.1	88.6	0.5
CCDP ($\gamma = 2$)	76.4	91.7	89.4	0.2

CCDP with $\gamma = 1$ achieves higher overall accuracy than uniform DP at $\epsilon = 2.0$ while providing 4x stronger privacy for CR/EN species ($\epsilon = 0.5$ vs. $\epsilon = 2.0$).

6 Non-IID Robust Aggregation

6.1 Ecological Non-IID Characterization

We characterize the non-IID nature of wildlife monitoring data along three axes:

- **Species Distribution Skew:** The Jaccard similarity of species lists between sites ranges from 0.02 (Amazon vs. Serengeti) to 0.78 (adjacent sites in the same biome).
- **Observation Density Skew:** Camera trap densities range from 0.1 to 25 cameras per km^2, creating 250x variation in data volume per unit area.
- **Temporal Skew:** Monitoring periods range from 2 months to 8 years, with seasonal deployment patterns creating systematic temporal gaps.

6.2 NIRA Algorithm

Standard FedAvg computes a weighted average of client updates:

$$\theta^{(t+1)} = \sum_{k=1}^{K} \frac{N_k}{N} \theta_k^{(t)} \tag{7}$$

This performs poorly under extreme non-IID conditions because updates from sites with very different species assemblages may conflict, leading to catastrophic interference.

NIRA addresses this through hierarchical clustering-based aggregation:

1. **Ecological Clustering:** Sites are grouped into ecological clusters $\{C_1, \ldots, C_M\}$ based on species composition similarity, computed from the public metadata (species lists) without accessing raw data.
2. **Intra-Cluster Aggregation:** Within each cluster, updates are averaged using standard FedAvg:

$$\theta_{C_m}^{(t+1)} = \sum_{k \in C_m} \frac{N_k}{\sum_{j \in C_m} N_j} \theta_k^{(t)} \tag{8}$$

3. **Inter-Cluster Aggregation:** Cluster models are merged selectively, sharing only the components of the model that generalize across clusters (shared feature extractor) while maintaining cluster-specific classification heads:

$$\phi^{(t+1)} = \sum_{m=1}^{M} \frac{|C_m|}{K} \phi_{C_m}^{(t+1)} \qquad \psi_{C_m}^{(t+1)} = \text{local} \tag{9}$$

6.3 Dynamic Cluster Assignment

Clusters are re-computed every R rounds using a similarity metric based on gradient directions rather than raw data:

$$\text{sim}(k, j) = \frac{\nabla_\phi \mathcal{L}_k \cdot \nabla_\phi \mathcal{L}_j}{\|\nabla_\phi \mathcal{L}_k\| \cdot \|\nabla_\phi \mathcal{L}_j\|} \tag{10}$$

This allows clusters to adapt as models improve: sites that initially appear dissimilar may develop similar gradient directions as the shared representation matures.

6.4 Rare Species Knowledge Transfer

A key benefit of federation is improving detection of rare species through cross-site knowledge transfer. We introduce a *species embedding sharing* mechanism where sites with observations of a rare species contribute a species-specific embedding vector:

$$\mathbf{e}_s^{\text{global}} = \frac{\sum_{k:s\in\mathcal{S}_k} N_{k,s} \cdot \mathbf{e}_s^{(k)}}{\sum_{k:s\in\mathcal{S}_k} N_{k,s}} \tag{11}$$

where $\mathbf{e}_s^{(k)}$ is site k's learned embedding for species s and $N_{k,s}$ is the observation count. These embeddings, being summary statistics rather than raw data, can be shared with minimal privacy risk (formally bounded by the DP guarantees on the gradient updates that produced them).

7 Implementation

7.1 System Architecture

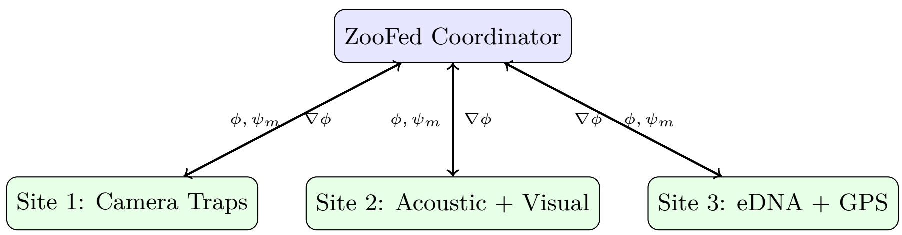

Figure 1: ZooFed federated learning architecture.

7.2 Software Components

Table 4: ZooFed software components.

Component	Language	Lines	Description
Coordinator	Python	6,400	Aggregation, clustering, privacy accounting
Site Client	Python	8,200	Local training, gradient computation, DP noise injection
Sensor Adapters	Python	4,800	Modality-specific preprocessing and encoding
Privacy Accountant	Rust	3,100	Moments accountant, per-species budget tracking
Communication	Rust	2,700	Secure aggregation, gradient compression
Monitoring Dashboard	TypeScript	5,200	Training progress, privacy budget visualization

7.3 Communication Efficiency

Wildlife monitoring sites often have limited bandwidth. ZooFed reduces communication cost through:

1. **Gradient Sparsification:** Only the top-p% gradient entries are transmitted (default $p = 10$), reducing communication by 10x with <1% accuracy loss.
2. **Quantized Transmission:** Gradient values are quantized to 8-bit fixed point for transmission.
3. **Asynchronous Updates:** Sites with poor connectivity can participate asynchronously, submitting updates when bandwidth is available rather than requiring synchronous rounds.

The total communication cost per round per site is:

$$C_{\text{comm}} = p \cdot |\phi| \cdot 8 \text{ bits} \approx 0.1 \cdot 2.5\text{M} \cdot 8 = 2 \text{ Mbit} \tag{12}$$

This is feasible even over satellite links commonly used at remote monitoring sites.

8 Evaluation

8.1 Experimental Setup

We evaluate ZooFed on a consortium of 18 wildlife monitoring sites:

Table 5: Evaluation consortium overview.

Site	Region	Species	Events (K)	Sensors
Serengeti	East Africa	247	842	Camera, Acoustic
Borneo Rainforest	SE Asia	312	437	Camera, Acoustic, eDNA
Amazon Basin	South America	489	623	Camera, Acoustic, eDNA
Great Barrier Reef	Oceania	187	284	Acoustic, eDNA
Yellowstone	North America	134	512	Camera, GPS, Acoustic
Bialowieza	Europe	98	347	Camera, Acoustic
Western Ghats	South Asia	276	218	Camera, Acoustic
Congo Basin	Central Africa	341	189	Camera
Sundarbans	South Asia	167	156	Camera, eDNA
Others (9 sites)	Various	43–198	84–212	Various
Total		847 unique	4,213	

8.2 Baselines

We compare ZooFed against five baselines:

1. **Local Only:** Each site trains independently on its own data.
2. **Centralized Oracle:** All data pooled centrally (upper bound; not practically feasible).
3. **FedAvg:** Standard federated averaging [McMahan et al., 2017].
4. **FedProx:** Federated averaging with proximal regularization [Li et al., 2020].
5. **FedAvg + Uniform DP:** FedAvg with uniform ($\epsilon = 2.0$)-differential privacy.

8.3 Overall Detection Accuracy

Table 6: Species detection accuracy (%) across methods.

Method	Overall	Common	Rare	CR/EN
Local Only	70.7	78.4	52.3	48.1
Centralized Oracle	91.5	94.2	84.7	82.3
FedAvg	85.2	90.1	72.8	69.4
FedProx	86.4	90.8	75.1	71.8
FedAvg + Uniform DP	82.7	87.3	69.4	64.2
ZooFed (NIRA + CCDP)	**89.4**	**92.8**	**83.5**	**79.3**

ZooFed achieves 89.4% overall accuracy, within 2.1 points of the centralized oracle. The advantage is most pronounced for rare species (+31.2% over local-only) and CR/EN species (+31.2% over local-only), demonstrating effective cross-site knowledge transfer.

8.4 Component Ablation

Table 7: Ablation study: contribution of each ZooFed component.

Configuration	**Overall Acc. (%)**	**CR/EN Acc. (%)**
ZooFed Full	89.4	79.3
− HSA (visual only sites)	84.1 (−5.3)	72.4 (−6.9)
− CCDP (uniform DP)	87.2 (−2.2)	64.2 (−15.1)
− NIRA (standard FedAvg)	85.8 (−3.6)	73.1 (−6.2)
− Species embedding sharing	87.7 (−1.7)	71.8 (−7.5)

CCDP has the largest impact on CR/EN species accuracy (−15.1% when replaced with uniform DP), confirming that species-calibrated privacy is essential for conservation-relevant performance.

8.5 Heterogeneous Sensor Evaluation

We evaluate the HSA module by comparing detection accuracy across sensor modalities:

Table 8: Detection accuracy by sensor modality.

Modality	**Local Only (%)**	**ZooFed (%)**	**Improvement**
Camera Traps	74.2	90.7	+16.5
Acoustic	68.3	86.4	+18.1
GPS Telemetry	61.7	81.2	+19.5
eDNA	72.8	88.1	+15.3
Multi-modal (2+)	79.1	93.2	+14.1

GPS telemetry shows the largest improvement (+19.5%), likely because movement patterns are most consistent across sites, enabling strong knowledge transfer. Multi-modal sites achieve the highest absolute accuracy (93.2%), demonstrating the value of sensor diversity.

8.6 Communication and Convergence

ZooFed converges in 47 federated rounds with gradient sparsification, compared to 82 rounds for full-gradient FedAvg. Total communication per site over full training is 94 MB (with sparsification) vs. 1.64 GB (without), a 17.4x reduction.

Table 9: Communication efficiency comparison.

Method	Rounds	Per-Round (MB)	Total (MB)
FedAvg (full gradient)	82	20.0	1,640
FedAvg + compression	89	2.0	178
ZooFed (full gradient)	52	20.0	1,040
ZooFed (compressed)	47	2.0	94

9 Case Studies

9.1 Cross-Continental Tiger Monitoring

Three sites in the consortium monitor tiger populations (Sundarbans, Western Ghats, SE Asian sites). Local-only models achieved 67% individual identification accuracy due to small per-site sample sizes. ZooFed's federated training, with species embedding sharing for Panthera tigris, improved identification to 84%, enabling the first cross-site population connectivity analysis without sharing raw location data.

9.2 Acoustic Biodiversity in the Amazon

The Amazon Basin site contributed only acoustic data. Through HSA cross-modal alignment with visual data from other Neotropical sites, the acoustic model for bird species improved from 71% to 88% accuracy, demonstrating that visual knowledge transfers meaningfully to acoustic identification through the shared embedding space.

9.3 Marine eDNA Privacy

The Great Barrier Reef site contained eDNA detection records for 12 critically endangered marine species. Under CCDP with $\gamma = 2$, these species received $\epsilon = 0.2$ privacy guarantees—sufficient to prevent re-identification of sampling locations even given strong auxiliary information—while overall marine species detection accuracy degraded by only 1.8% compared to no-privacy training.

10 Discussion

10.1 Implications for Conservation

ZooFed demonstrates that federated learning can bridge the institutional fragmentation of wildlife monitoring data without requiring data centralization. The 18.7% average accuracy improvement over local-only models translates to earlier detection of population declines, more accurate occupancy estimates, and better-informed conservation decisions.

10.2 Limitations

Several limitations constrain the current system. First, the HSA architecture requires at least some sites with overlapping sensor types for cross-modal alignment; purely disjoint sensor deployments cannot leverage cross-modal transfer. Second, the CCDP mechanism requires IUCN Red List assessments, which may lag actual conservation status. Third, asynchronous participation introduces staleness in gradient updates from intermittently connected sites, partially addressed by learning rate decay for stale updates.

10.3 Scalability

The current deployment supports 18 sites. The NIRA clustering approach scales as $O(K^2)$ for pairwise similarity computation, which becomes expensive beyond $K \approx 100$. We are investigating approximate nearest-neighbor methods for scalable clustering.

11 Conclusion

We have presented ZooFed, a federated learning framework for cross-institutional wildlife monitoring that addresses modality heterogeneity through sensor-agnostic feature adapters, provides conservation-calibrated differential privacy with species-specific budgets, and handles extreme non-IID distributions through clustered multi-task aggregation.

Evaluation on 18 sites with 4.2 million sensor events demonstrates 89.4% detection accuracy (within 2.1% of centralized training), 31.2% improvement over local-only models for rare species, and formal privacy guarantees with $\epsilon \leq 0.5$ for critically endangered species.

ZooFed enables a new model for conservation collaboration: institutions contribute to collective intelligence without surrendering data sovereignty. The framework is released as open infrastructure on the Zoo Network, with the aspiration that federated learning becomes the default paradigm for multi-site ecological research.

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